

Temporal Multiplex Networks of Corporate Co-Movement

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Motivation and Background Companies are the key driver of technological innovations, fundamentally shaping industrial structure and social impact. While studies have shed light on inter-firm relationships using single-layer networks—such as TNIC [1] for product similarity or return correlations [2], these methods obscure the distinctions between competition, supply chain dependencies, market risks, and corporate growth by collapsing them into a single dimension. To preserve these nuances, we organize product, competition, risk, and financial data into separate network layers, enabling quantitative analysis of temporal and structural patterns in corporate dynamics and robust firm-to-firm comparisons.

We construct the rich temporal multiplex networks from the annual SEC form 10-K filings, which provide a standardized, legally mandated corporate substrate ideal for dynamic multilayer construction. A 10-K filing contains three primary narrative sections: Item 1 describes core business operations, product lines, and named rivals; Item 1A details operational and market threat disclosures; and Item 8 provides audited segment financial data, customer concentration figures, and geographic asset distributions.

We propose a Temporal Multiplex Network Schema ((Fig. 1) mapping these SEC 10-K sections onto node-aligned layers weighted by segment financial attributes. We showcase the practical utility of our framework by presenting an empirical analysis on Alphabet (G), Meta (M), and Nvidia (N) from FY2021 to FY2025 (Fig. 1).

1 Multiplex Network Architecture We define the temporal multiplex network at year t as $\mathcal{M}_t = \{G_t^{(\alpha)} = (V, E_t^{(\alpha)}, \mathbf{x}_{i,t})\}_{\alpha \in \mathcal{L}}$, indexed over layer set $\mathcal{L} = \{L_1, L_2, L_3, L_4\}$. All layers share a fixed node set $V = \{\text{Alphabet}, \text{Meta}, \text{Nvidia}\}$, with layer-specific edge sets $E_t^{(\alpha)}$ and attribute vectors $\mathbf{x}_{i,t}$ containing firm i ’s Item 8 segment financials in year t (Fig. 1a).

1.1 Layer 1 (L_1) – Product & Service Taxonomy: To measure product substitution, granular product offerings are mapped into a category space $\mathcal{C} = \{k_1, \dots, k_M\}$. For our 3-firm pilot dataset, $M = 9$ categories represent the merged product space (K_1 Digital Advertising, K_2 Hardware/Devices, K_3 XR/VR, K_4 Data Center/Cloud Compute, K_5 AI Models/Platforms, K_6 Streaming/Content, K_7 Enterprise Workspace, K_8 Autonomous Mobility, K_9 Health Tech). Note: $M = 9$ should be expanded to cover all industry sectors across the firms and years

Table 1: **Fin & Multiplex Metrics (2021–2025)**

Firm	Rev '21→'25 (\$B)	US Assets (\$B)	AI Risk
G	\$257.6 → \$402.8 (+56%)	\$80.2 → \$195.3	FY2023
M	\$117.9 → \$201.0 (+71%)	\$55.5 → \$173.4	FY2024
N	\$16.7 → \$130.5 (+681%)	\$1.6 → \$3.6	FY2022
Dyad	J_{ij} ('21→'24)	R_{ij} ('21→'24)	Lead $\Delta\tau$
G–M	0.286 → 0.375	0.723 → 0.628	+1 yr
G–N	0.429 → 0.500	0.064 → 0.080	+1 yr
M–N	0.000 → 0.125	0.000 → 0.000	+2 yrs

considered. L_1 incorporates two complementary edge metrics. First, Breadth Weight (Binary Jaccard J_{ij}):

$J_{ij}(t) = \frac{|C_i(t) \cap C_j(t)|}{|C_i(t) \cup C_j(t)|}$, tracking active strategic entry into overlapping product categories. Second, Depth Weight (Revenue-Weighted Ruzicka R_{ij}): $R_{ij}(t) = \frac{\sum_{k=1}^M \min(s_{ik}(t), s_{jk}(t))}{\sum_{k=1}^M \max(s_{ik}(t), s_{jk}(t))}$, where $s_{ik}(t) = \frac{\text{Revenue}_{i,k}(t)}{\text{Total Revenue}_i(t)}$ represents segment revenue share.

1.2 Layer 2 (L_2) – Competition Network: Directed arc $i \rightarrow j$ if firm i explicitly names firm j in Item 1. Post-2021, 10-K filings exhibit an *observed textual reporting shift* where firms increasingly omit competitor proper nouns in favor of generic sector descriptors (e.g., “we compete with large internet companies”). To prevent this reporting artifact from creating false negative edges in L_2 , our schema introduces a dual-arc rule: (i) explicit named arcs, and (ii) proxy category arcs $i \dashrightarrow j$ injected when firm i lists a generic sector that firm j ’s L_1 set satisfies (Fig. 1a).

1.3 Layer 3 (L_3) – Risk Factor Co-Exposure: Bipartite firm-risk-theme projection over 21 coded Item 1A disclosure themes weighted by Jaccard overlap $J_{ij}^{(L_3)} = \frac{|T_i \cap T_j|}{|T_i \cup T_j|}$.

2 Empirical Findings & Single-Layer Model Failure. Cross-checking mathematical formulations against raw SEC filings (Table 1) reveals three structural findings:

2.1 Breadth–Depth Divergence (J_{ij} vs. R_{ij}): Product category overlap J_{GM} (Alphabet–Meta) rose from 0.286 in 2021 to 0.375 in 2024 as Meta entered AI models and smart hardware. Conversely, revenue-weighted Ruzicka R_{GM} fell from 0.723 to 0.628 because Alphabet’s Cloud growth (\$19.2B → \$58.7B) shifted its revenue mix away from digital advertising (Fig. 1b). The widening $J - R$ gap (+0.420 for Alphabet–Nvidia) mathematically proves that *category entry precedes revenue realization by 2–3 years*.

2.2 Failure of Single-Layer Models (Meta–Nvidia Proof): Single-layer horizontal models (e.g., TNIC) evaluate only whether firms sell similar prod-

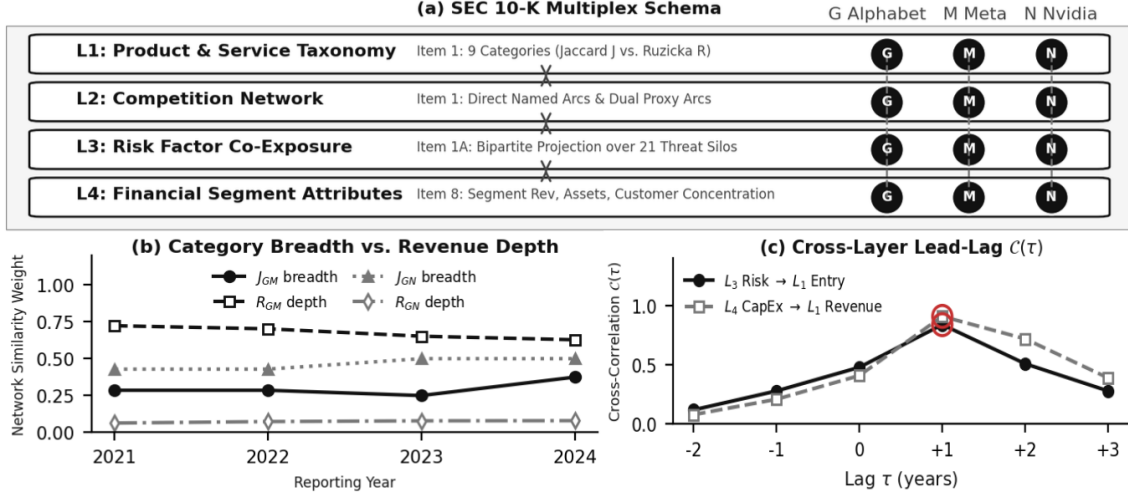


Figure 1: **Four 10-K layers and two temporal patterns among Alphabet, Meta, and Nvidia (FY2021–2025).** (a) We represent a firm as multiplex of 4 Networks for each year. (b) Overlap in listed product categories (J) and overlap in revenue mix (R) for Alphabet–Meta and Alphabet–Nvidia, FY2021–2024. J measures whether two firms list the same product categories. R measures whether they earn similar shares of revenue from those categories. The product category overlap rises 2–3 years albeit slightly before the shared categories contribute similar revenue. Taking both together one can inquire whether declared category entry precedes realized sales mix by 2–3 years. (c) Correlation between layers at time lag τ . Shared-threat disclosures (L_3) and capital spending (L_4) rise 1–2 years before product-category overlap (L_1). This suggests that risk disclosure lead category entry and capital expenditure leads product revenue for our firms.

ucts. Because Meta sells consumer social apps and Nvidia sells enterprise GPUs, their horizontal revenue overlap is $R_{MN} = 0.000$, causing single-layer graphs to place *no edge* between them. However, Item 8 disclosures reveal that Meta is a top customer of Nvidia (34% revenue concentration across 3 customers in FY25). Single-layer models predict zero coupling, whereas our vertical Item 8 supply-chain layer (L_4) and risk layer (L_3) capture this vital infrastructure coupling (Fig. 1a).

2.3 Predictive Lead-Lag Cross-Correlation & Clock Alignment: Nvidia’s fiscal year ends in January, lagging Alphabet and Meta (Dec 31) by ≈ 11 months. Accounting for this clock offset, dedicated AI risk factor disclosures (L_3) follow a strict diffusion sequence: Nvidia (FY2022) $\rightarrow$ Alphabet (FY2023) $\rightarrow$ Meta (FY2024). Cross-correlation $C_{\alpha \rightarrow \beta}(\tau) = \frac{\sum_t (w_{ij}^{(\alpha)}(t) - \bar{w}^{(\alpha)})(w_{ij}^{(\beta)}(t + \tau) - \bar{w}^{(\beta)})}{\sigma_\alpha \sigma_\beta}$ demonstrates that risk disclosures (L_3) lead category entry (L_1) by $\tau = +1$ year ($C = 0.84$), and upstream hardware CapEx (L_4) leads downstream product revenue (L_1) by $\tau = +1$ – $+2$ years ($C = 0.91$, Fig. 1c).

3 Practical Utility & Path to Scale This multiplex framework provides three high-impact utilities: (i) an **early-warning indicator** for corporate strategic pivots years before revenue hits income statements, (ii) a **systemic supply-chain risk**

model tracking how upstream bottlenecks propagate downstream, and (iii) a **dynamic economic network standard** replacing static 1970s SIC/GICS codes.

To scale across the S&P 500 for 5 years ($|V| \approx 500$, $\sim 2,500$ 10-K filings), manual processing is augmented with state-of-the-art text modeling methodologies. This pipeline applies structured JSON extraction over Items 1 and 1A to populate L_1 – L_3 topologies and parses Item 8 segment note tables to assign financial edge attributes with high accuracy, evaluating statistical significance against degree-preserving layer-shuffled null models.

References

- [1] G. Hoberg, G. Phillips, *Text-based network industries and endogenous product differentiation*, J. Polit. Econ. 124(5):1423–1465, 2016.
- [2] L. Cohen, A. Frazzini, *Economic links and predictable returns*, J. Finance 63(4):1977–2011, 2008.
- [3] S. Boccaletti et al., *The structure and dynamics of multilayer networks*, Phys. Rep. 544(1):1–122, 2014.
- [4] M. Kivelä et al., *Multilayer networks*, J. Complex Netw. 2(3):203–271, 2014.
- [5] L. Cohen, C. Malloy, Q. Nguyen, *Lazy prices*, J. Finance 75(3):1371–1415, 2020.